# Homework 1 - Multivariate Analysis

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## 1. Exploratory Data Analysis

### 1a. Data Import

We imported the dataset into the R environment. We discarded the initial variable 'No', which served merely as a row index, from the data frame because it holds no predictive or analytical value and would skew variance and distance calculations in the subsequent multivariate analysis. Following the cleanup, the dataset consists of 71 rows and 21 columns.

### 1b. Convert Minutes

We split the variable "Min" using strsplit(data$Min, ":") to extract minutes and to convert it to a numeric variable.

### 1c. Set Player Names as Row Labels

We set the player names as row labels, so that we can identify individuals directly in the PCA and MDS plots.We used make.unique() as a safeguard against duplicate names. We then removed "PLAYER" and "Min", because they are now redundant with the row labels and "Min 2".

## 1d. Filter Players

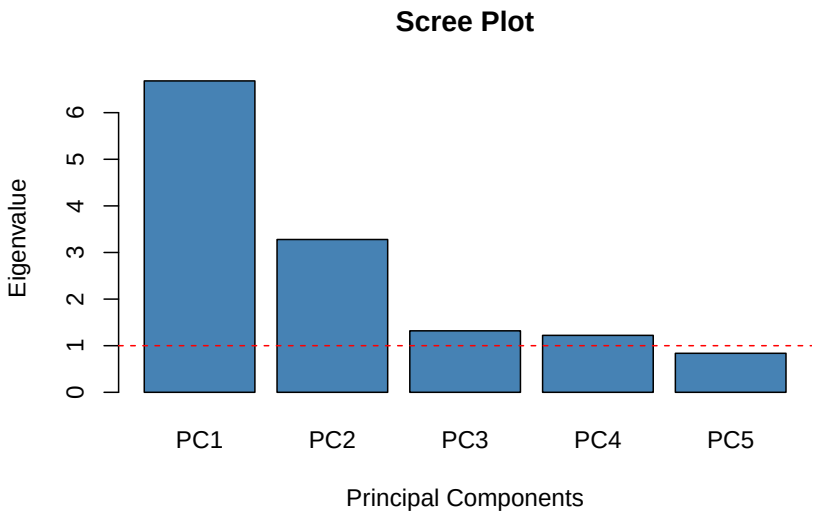
We only keep players that at least played for 5 minutes or more than in 5 games. Their averages come from very little playing time, so they are unstable and could distort our further analysis. After filtering, 60 of the original 71 players remained.

## 2. Principal Component Analysis (PCA)

### 2a

We ran PCA on all numerical variables, standardised (scale.unit = TRUE). Standardising was necessary because the variables are on different scales: percentages, counts, and minutes. We included TEAM and POSITION as supplementary qualitative variables and PIR as a supplementary quantitative variable. They do not influence the construction of the components but are projected onto them afterwards to help interpretation. We excluded PIR from the active variables because it is itself a composite of the other statistics.

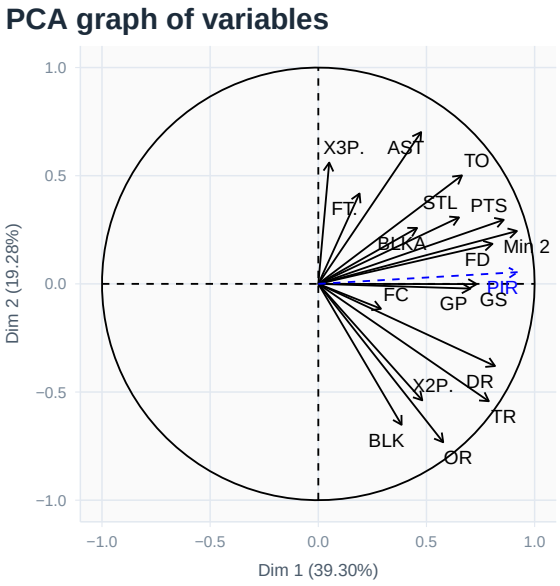
## 2b. Number of Principal Components

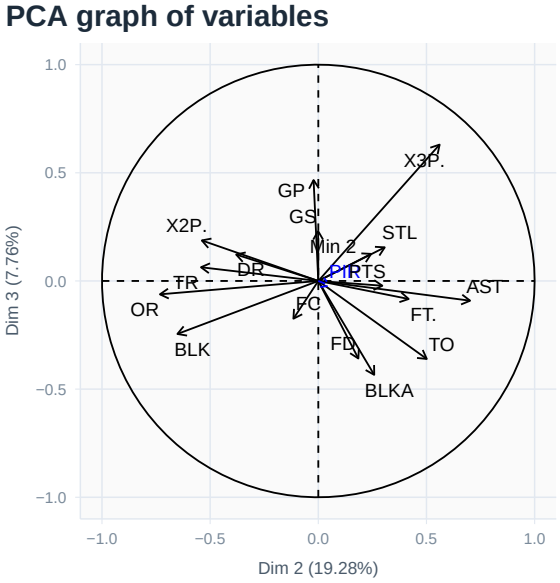



Based on the Kaiser criterion, four principal components should be retained, as their eigenvalues exceed 1. The scree plot shows a particularly steep decline after PC2, indicating that the first two components explain most of the variance (cumulative 58,58%). However, we retain the first 3 PCs (PC3 explaining 7,7% of variance), explaining a cumulative variance of 66,34%, since PC3 brings valuable information to the analysis by capturing the relationship between three-point shooting vs. physical play that we explain further in the following questions.

## 2c

The variable plots show the correlation of each active variable with the principal components. Longer arrows indicate variables that are better represented in the plane shown. PIR appears as a dashed supplementary arrow.



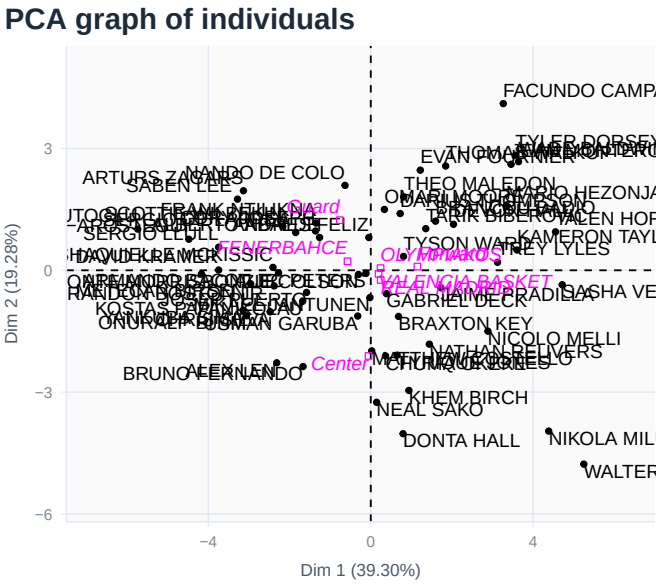




## 2d

**PC1 (39.30%) – Overall Player Activity:** Most statistical variables are positively correlated with PC1, particularly points, minutes played, and rebounds. This dimension represents overall player activity and statistical contribution.

**PC2 (19.28%) – Playmaking vs. Rebounding:** Assists, turnovers, and steals correlate positively with PC2, while rebounds and blocks correlate negatively. This dimension distinguishes playmaking-oriented players from inside players.

**PC3 (7.76%) – Three-Point Shooting vs. Physical Play:** Three-point shooting percentage and games played correlate positively with PC3, while blocks against and turnovers correlate negatively. This dimension may reflect differences in playing style, although its interpretation is less clear.







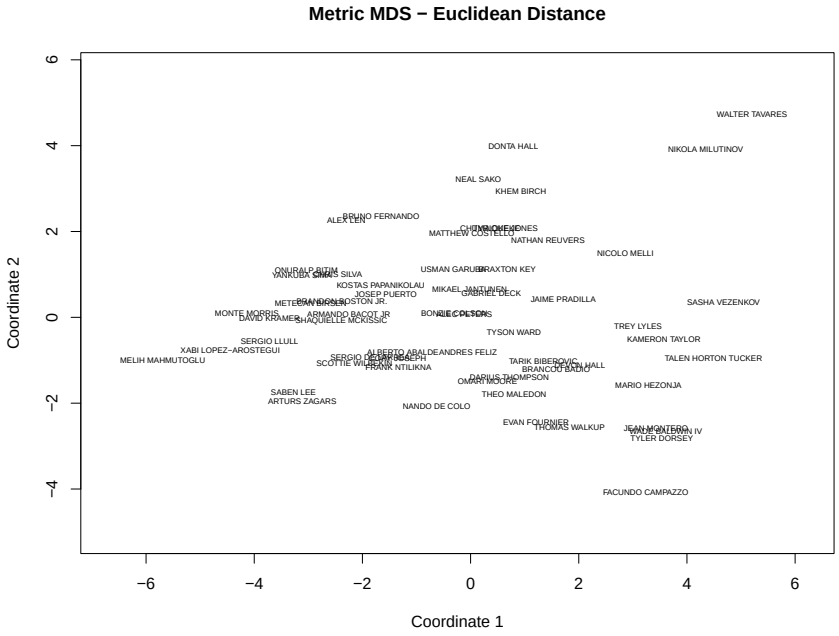
The PC1–PC2 plot (58.58% explained variance) separates players mainly by overall statistical activity (PC1) and playing style (PC2). Players 33 and 14 show high PC1 scores, while player 44 has a low score. PC2 distinguishes playmaking-oriented players, such as player 24, from more inside-oriented players, such as players 33 and 17.

The PC2–PC3 plot (27.04% explained variance) highlights additional differences in playing style, particularly three-point shooting. Players 25 and 23 have relatively high PC3 scores. Player positions partly reflect their roles, with guards tending towards higher PC2 values and centers towards lower values. However, the teams overlap considerably, indicating no clear separation by team.

# 3. Multidimensional Scaling (MDS)

## 3a. Metric MDS - Euclidean Distance

We applied classical (metric) MDS to the Euclidean distance matrix of the standardised numerical variables (PIR excluded, as in PCA). Classical MDS on Euclidean distances of standardised data gives the same configuration as the PCA individual map, so the Euclidean MDS plot is essentially the PCA PC1–PC2 individual plot, labelled with player names.

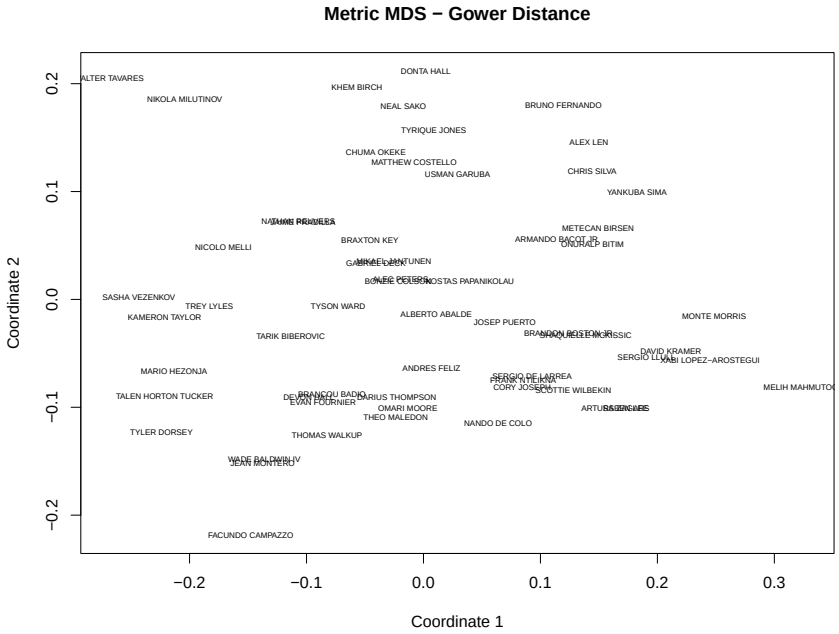



## 3b

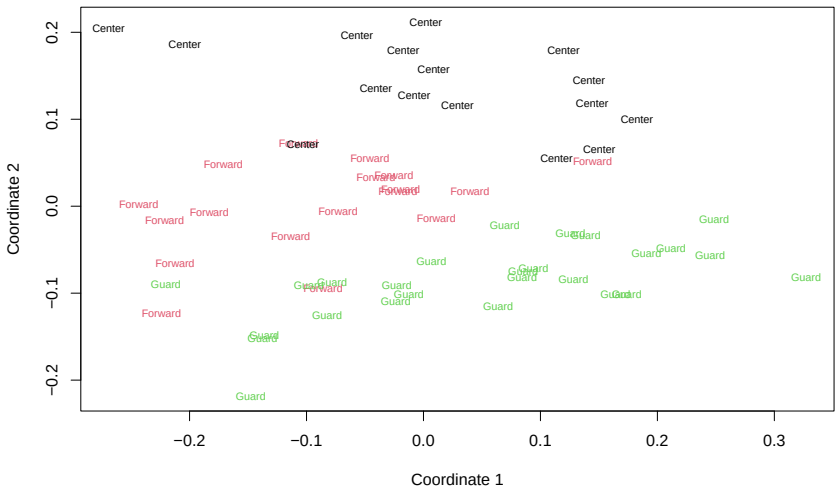
The Euclidean MDS plot shows a central concentration of players with relatively similar statistical profiles, without clearly separated clusters. Some players, such as Walter Tavares, Nikola Milutinov, and Facundo Campazzo, occupy more isolated positions, indicating distinct playing characteristics. Overall, the plot reveals both similarities among many players and substantial differences for certain individuals.

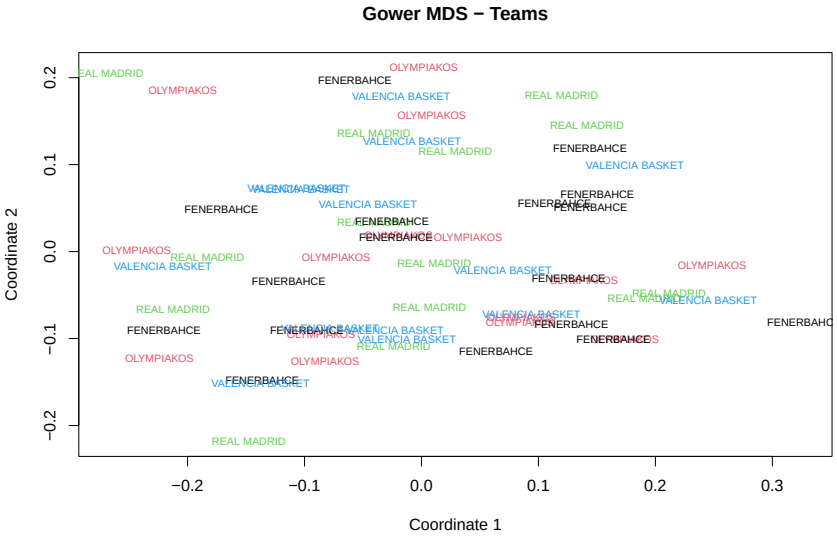
## 3c. Metric MDS - Gower Distance

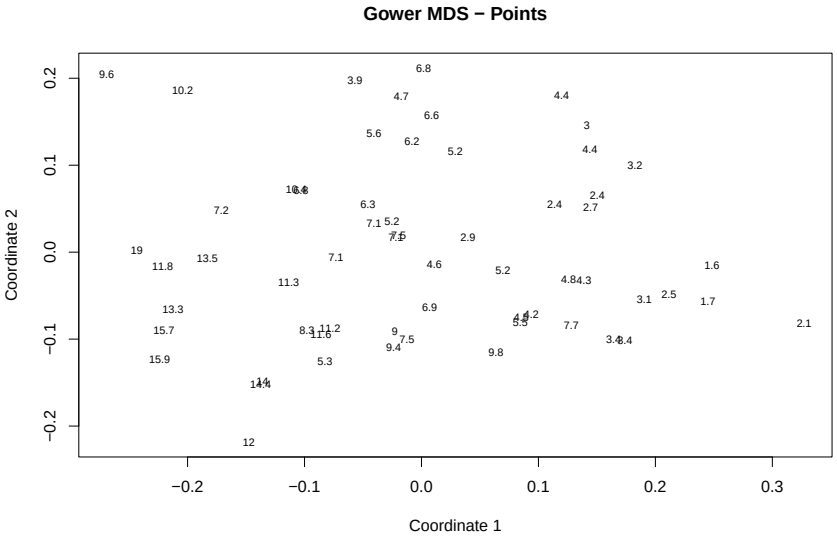
We used Gower distance because it can combine numerical variables with the categorical variable POSITION. Each variable contributes a dissimilarity between 0 and 1: range-scaled absolute differences for numerical variables, and simple mismatch for categorical ones. We then applied Metric MDS to the resulting distance matrix. The first two coordinates explain 57,63% of the total variation.

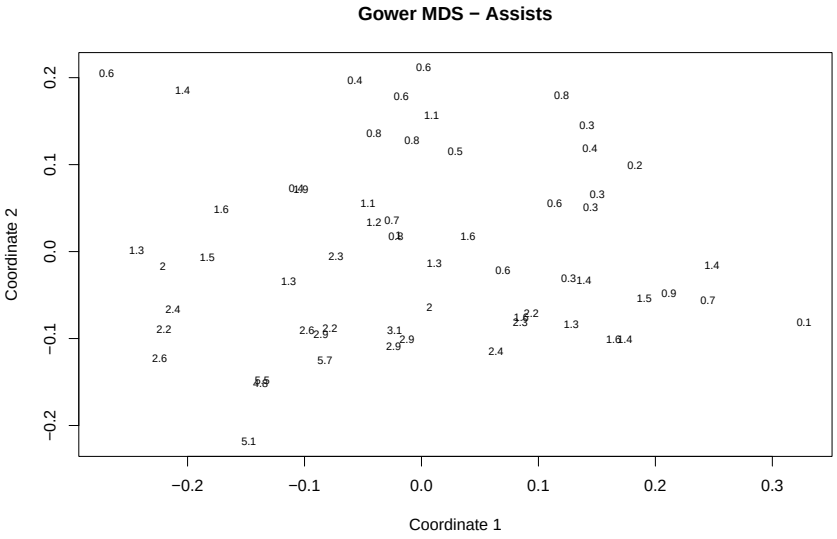



# 3d. Interpretation of Gower MDS Dimensions




Gower MDS − Teams
Coordinate 2
Coordinate 1
−0.2
−0.1
0.0
0.1
0.2
0.3
OLYMPIAKOS
FENERBAHCE
VALENCIA BASKET
REAL MADRID


Gower MDS − Points
Coordinate 2
Coordinate 1
−0.2
−0.1
0.0
0.1
0.2
0.3




**Coordinate 1 – Scoring Activity:** Players with higher average points generally appear on the left, while lower-scoring players tend to appear on the right, suggesting that this dimension partly reflects scoring activity.

**Coordinate 2 – Playing Position and Playmaking:** Centers are mainly located at positive values, guards at negative values, and forwards in between. Higher assist averages also tend to occur at negative values, indicating that this dimension distinguishes inside-oriented players from playmakers.

**Team Membership:** Players from all four teams overlap considerably, suggesting that team membership does not strongly explain the MDS dimensions.

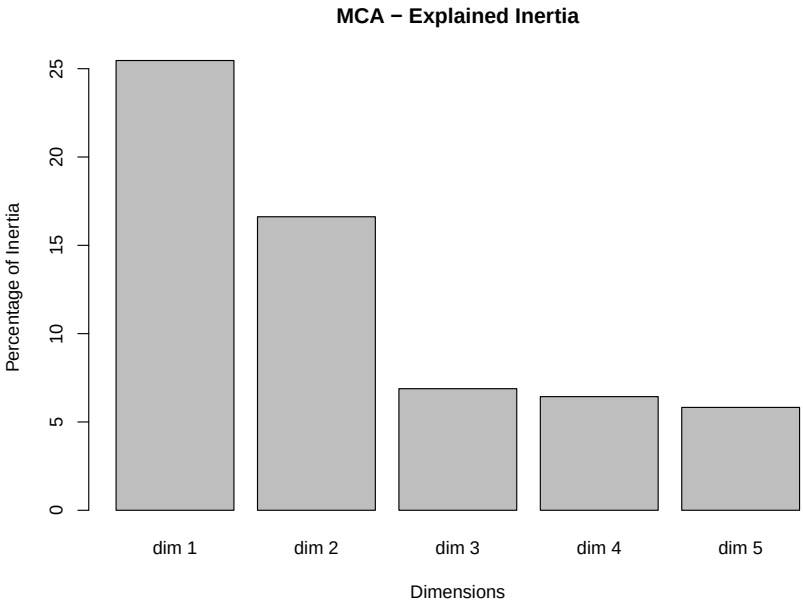
## 3e

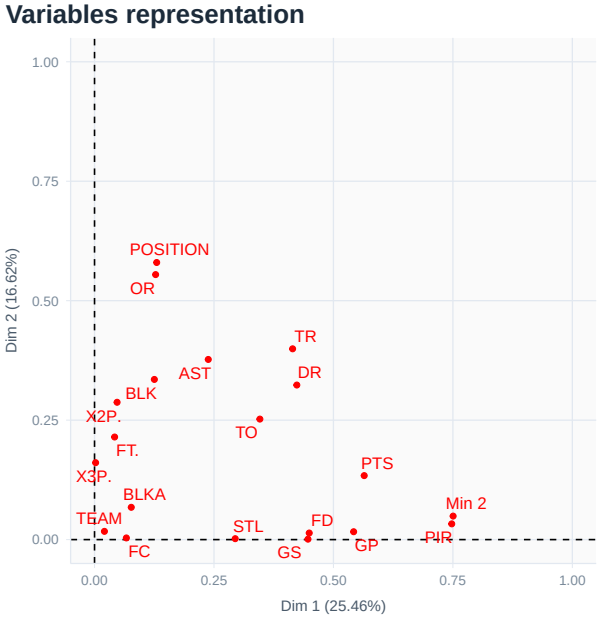
Gower MDS provides clearer grouping of players by position, with centers, guards, and forwards showing a more distinct separation. This is because Gower distance incorporates the categorical variable POSITION alongside numerical statistics. Euclidean MDS mainly reflects differences in numerical performance and shows less distinct groups. # 4. Multiple Correspondence Analysis (MCA)

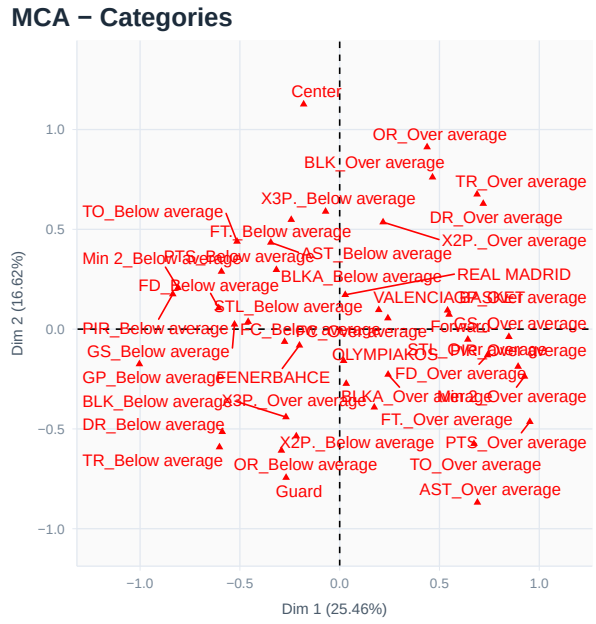
We converted each numerical variable into a two-level factor: "Over average" if the player's value exceeds the variable mean, and "Below average" otherwise. This way we are able to analyse the continuous statistics together with TEAM and POSITION in MCA. The trade-off was a loss of information, since we discard the magnitude of differences within each category.

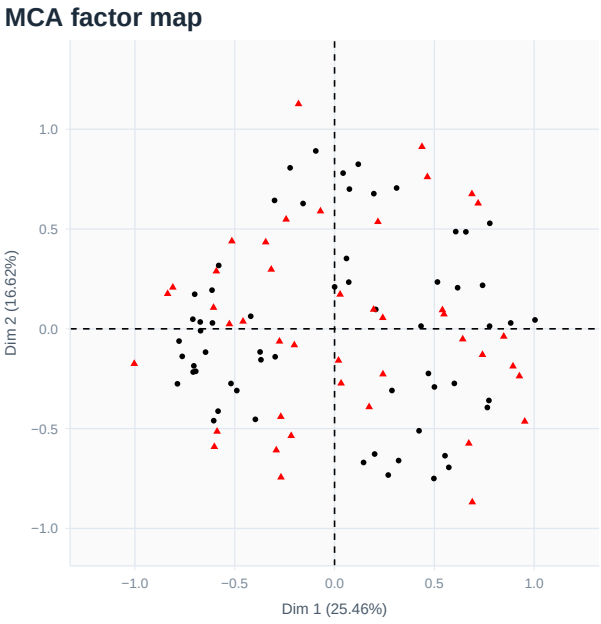
## 4a. Convert Numerical Variables to Binary Categories

## 4b. Multiple Correspondence Analysis






The first two MCA dimensions explain **42.08% of the total inertia** (25.46% and 16.62%).

**Dimension 1 – Overall Statistical Involvement:** This dimension is mainly associated with minutes played ($R^2 = 0.750$), PIR ($R^2 = 0.748$), and points scored ($R^2 = 0.564$). Above-average performance categories are located on the positive side, while below-average categories appear on the negative side.

**Dimension 2 – Inside Play vs. Playmaking:** Playing position ($R^2 = 0.580$), offensive rebounds ($R^2 = 0.555$), and assists ($R^2 = 0.377$) strongly characterize this dimension. Centers and above-average rebounding categories are located at positive values, while guards and above-average assists appear at negative values.

The individual factor map shows differences in player profiles without clearly separated clusters. Team categories are located near the origin, suggesting little influence on the first two dimensions. However, Dimension 3 (6.88%) is strongly associated with team membership ($R^2 = 0.719$).

### 4c

All three methods reveal similar patterns. PCA and MCA identify overall statistical involvement as the first dimension and differences between playmaking and inside-oriented players as the second. Euclidean MDS highlights differences in numerical player statistics, while Gower MDS provides clearer separation by playing position. Unlike PCA, MCA uses categorized performance variables, resulting in some information loss but making above- and below-average performance profiles easier to interpret.